

Monte Carlo Counterparty Credit Risk Engine

Executive summary: what the risk is, how we measure it, what we found, and how far to trust it

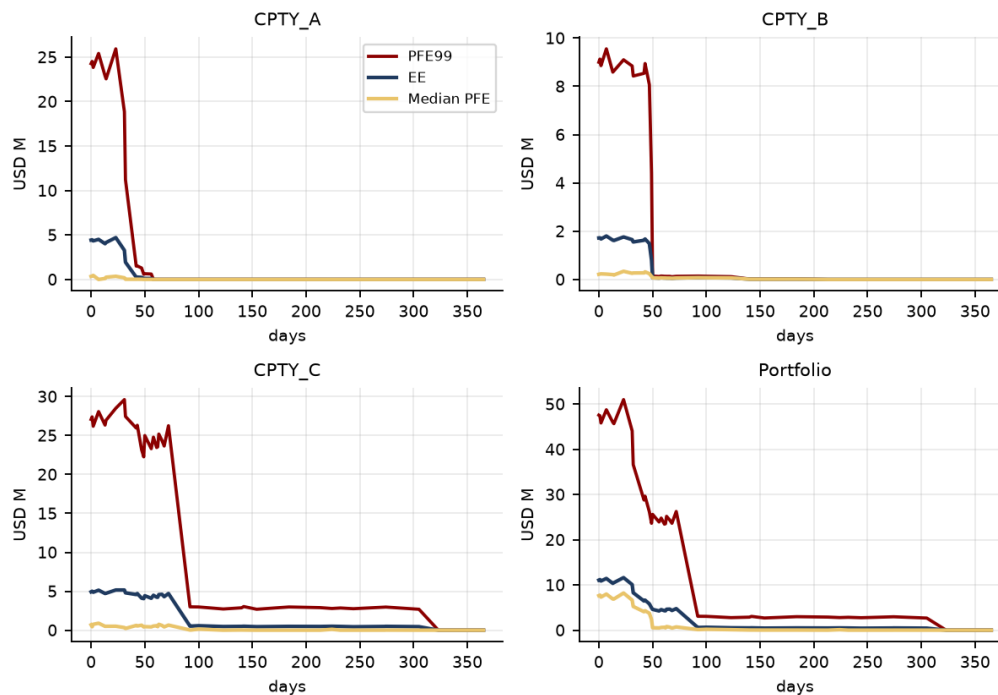
ESF derivatives book | 16 trades | 3 counterparties | exposure on the brief's 10-day close-out definition | full detail in [Capitolis_CCR_Complete_Report.pdf](#)

The answer in one page

\$51M	\$11.6M	\$8.2M	78%
portfolio MPE99, close-out definition (2026-09-20)	peak expected exposure (EE)	peak median PFE	portfolio MPE as a share of the sum of the three counterparties

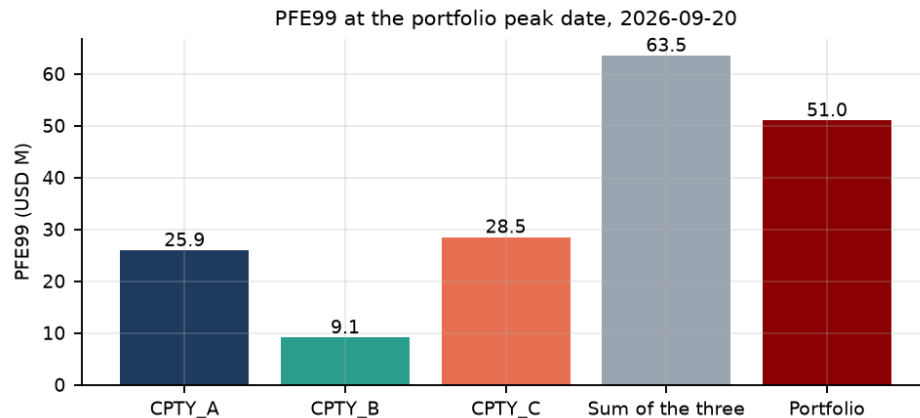
- Exposure follows the brief: the move of the netted value from its prior-day level over a 10-business-day close-out, no initial margin, per trade and per counterparty, one-year horizon.
- CPTY_A and CPTY_B are equity risk (pay-equity swaps: we gain when shares fall); CPTY_C is mostly interest-rate risk through one \$500M short forward on a 2049 Treasury (DV01 \$594k per bp) with an equity book beside it.
- Counterparty MPE99: A \$25.9M, B \$9.6M, C \$29.6M. The portfolio figure \$51.0M is 78% of their sum: netting does not cross counterparties and the books peak within about four weeks of each other.
- Credit: CVA \$12k on the margined close-out exposure and \$247k if no margin were ever called (BBB proxy); SA-CVA capital \$0.10M; SA-CCR exposure at default \$260.3M.
- Confirming CSA terms is the most valuable open question: it decides which of the two exposure definitions applies.

What we measure



$\text{exposure}(t) = \max(V(t + 10bd) - V(t - 1bd), 0)$: variation margin is the prior-day value and stops at default, so only the 10-day move is at risk. EE is the average, the median PFE the typical case, PFE99 the level exceeded in 1 of 100 scenarios, MPE the peak of PFE.

Why the portfolio MPE is close to the sum of its parts



- The portfolio exposure is a sum of per-counterparty positive parts; nothing nets across counterparties. Its 99th percentile is below the sum of the standalone ones only where the tails do not coincide.
- Exposures are positively dependent (rank correlation A-B +0.34, A-C +0.33, B-C +0.12): every netting set holds pay-equity swaps, so all gain when equities fall; CPTY_C is not a pure rate set.
- Bonds and equities: in the last three years yield changes and stock returns are essentially uncorrelated (mean +0.00). The book is short both, so a flight to quality (yields fall with equities) makes the gains oppose each other and both falling together makes them coincide: portfolio MPE99 -15% and +18% against the base (report, Section 7.2).

The book: 16 trades, short-dated

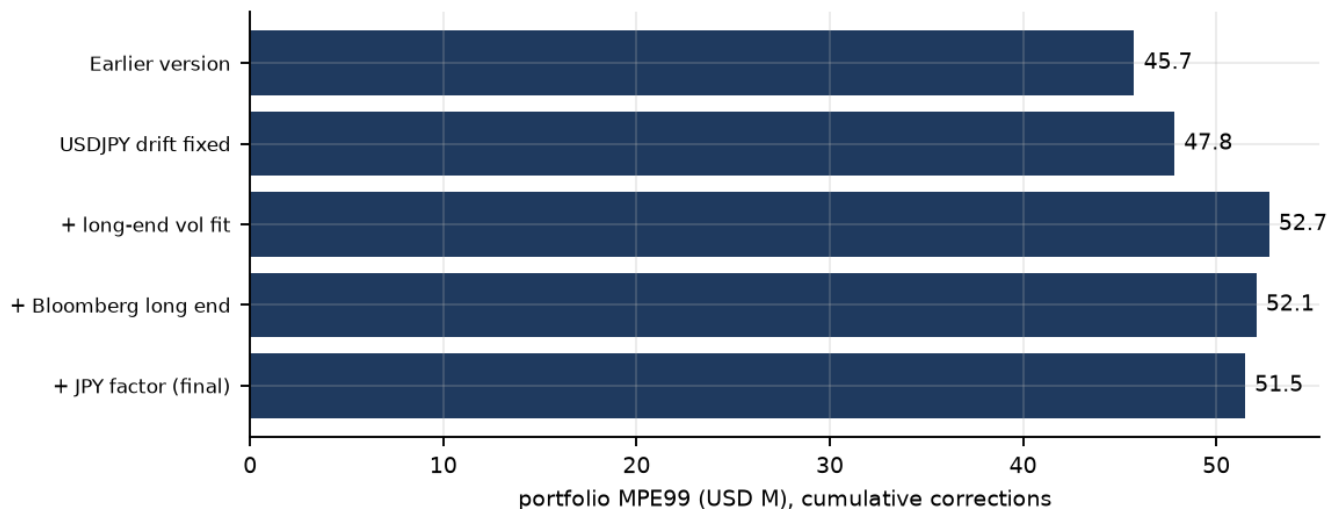


8 equity total return swaps (incl. JPY compo), 4 bond forwards, 4 bond TRS across three counterparties. All priced with the supplied, independently reviewed pricer library.

How the engine works

Step	What we do	Choice
1 Calibrate	SOFR curve from CME futures with the Bloomberg long end spliced on; realised vols; USD rate sigma fitted to Treasury yield vols; 40x40 correlation	Real market data throughout
2 Simulate	USD rate: Hull-White one-factor (G2++ two-factor compared). JPY rate: second Hull-White factor. Equities and USDJPY: correlated GBM driven by the rates	Exact fit to today's curve; negative-rate capable; drifts from the martingale condition
3 Sample	Latin-Hypercube scenarios (5,000 for reporting); Cholesky for correlation (PCA factor model optional)	Lowest measured error of 5 methods
4 Dates	Market pillar dates plus every trade's event dates, with $t - 1\text{bd}$ and $t + 10\text{bd}$ nodes	Exact cash-flow dates and the brief's close-out
5 Reprice	Every trade re-priced in every scenario and node with the standard pricers	No new pricing logic
6 Aggregate	Close-out exposure, netted by counterparty and per trade: EE, median PFE, PFE99, MPE	Uncollateralized level exposure alongside

What changed since the earlier version, and why



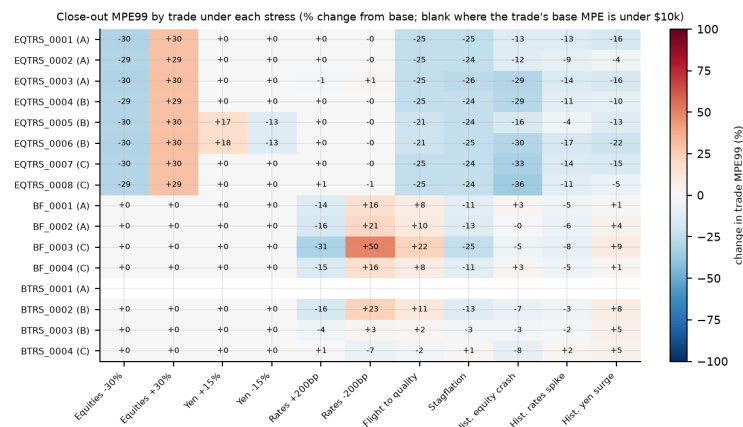
- Four corrections, each introduced on the same random numbers: USDJPY drift sign (found via a martingale test), volatility fitted to long-end yield moves (the largest effect, on CPTY_C), Bloomberg long end on the USD curve (BF_0003 \$101M to \$121M), JPY factor wired in.
- The long-dated bond forward drives the change: CPTY_C's MPE99 rises most. Details and per-counterparty numbers in the report, Section 7.3.

Model risk: which inputs matter (change in MPE99)

Perturbation (one at a time)	CPTY_A	CPTY_B	CPTY_C	Portfolio
USD rate sigma x1.25	+3%	+1%	+24%	+9%
Mean reversion x3	+2%	+1%	-13%	-5%
Equity/FX vols x1.25	+29%	+26%	+11%	+19%
Equity/FX vols x2	+98%	+95%	+32%	+71%
Equity correlations up	+23%	+35%	+9%	+22%
Yields fall with equities	+1%	+1%	-11%	-15%
Yields rise as equities fall	+3%	+1%	+21%	+18%
Two-factor G2++	-1%	-3%	-0%	-3%

Same random numbers, 2,000 scenarios. Equity volatility and the rate-equity dependence move the portfolio number most; the choice of one or two rate factors matters least.

Stress testing: 11 scenarios, same random numbers, every trade, every date



- Data: 37 names, USDJPY, Treasury yields 2014-2026; 8 round-number shocks and 3 historical 10-day windows chosen by rule; instantaneous shocks, base-model volatilities.
- Close-out MPE99 moves at most -24% (Stagflation); the uncollateralized level MPE99 +86% (Stagflation): the margined measure is far less sensitive to level shocks.
- Sensitivity is product specific (heat map): equity swaps react to equity and FX shocks, the bond forwards and bond TRS to rate shocks.

Backtest: realised 10-day moves against the model's quantiles

Netting set (equity and FX leg)	99%: exceptions / expected	Kupiec p-value	95%: mean exception rate over 10 offsets
CPTY_A	4 of 242 / 2.4	0.35	3.0% (nominal 5%)
CPTY_B	6 of 242 / 2.4	0.05	5.3% (nominal 5%)
CPTY_C	2 of 242 / 2.4	0.78	3.2% (nominal 5%)

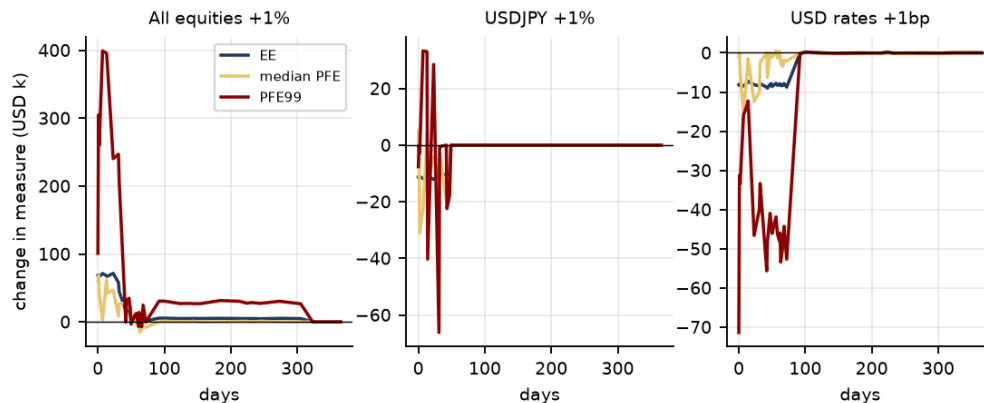
- Out of sample: at every as-of date the model is calibrated on the previous 3 years only; non-overlapping 10-day windows; Kupiec proportion-of-failures test.
- Equity and FX leg: the 99% quantile is close to calibrated (CPTY_B slightly light-tailed, as GBM implies); the 95% quantile is conservative for A and C.
- Rates leg: the long-end fit of the USD rate volatility gives 5.7% / 1.5% exception rates at 95% / 99% (nominal 5% / 1%); the earlier overnight-SOFR calibration is regime dependent (63bp now against 96bp for the fit).

CVA walk-through, DVA, FVA and capital

	CPTY_A	CPTY_B	CPTY_C	Total
Close-out exposure: CVA	\$2k	\$1k	\$8k	\$12k
Close-out exposure: DVA	\$2k	\$1k	\$8k	\$11k
Close-out exposure: FVA	\$0k	\$0k	\$0k	\$1k
Uncollateralized level exposure: CVA	\$7k	\$4k	\$237k	\$247k
Uncollateralized level exposure: DVA	\$3k	\$2k	\$3k	\$8k
Uncollateralized level exposure: FVA	\$4k	\$2k	\$234k	\$240k

- $CVA = LGD \times \text{sum of } 0.5 (DEE(t-1) + DEE(t)) \times \text{default probability}$, where DEE is the expected discounted exposure and the default probability comes from a BBB bond-spread proxy (60% LGD). The margined close-out CVA is small; the level CVA is the price if no margin were ever called.
- DVA and FVA rest on assumed own credit (BBB) and funding spread; total that avoids double counting = $CVA - DVA + FCA$.
- SA-CVA capital: \$0.10M on the close-out exposure and \$2.23M on the uncollateralized exposure; SA-CCR EAD \$260.3M (dominated by the replacement cost of BF_0003).
- For the next session: which exposure definition prices CVA, real ratings or CDS, own credit, wrong-way risk, capital approach.

Greeks: how they are generated and what they cost



- Method: bump-and-reprice with common random numbers (same draws in base and bumped runs); central differences for delta and gamma; equity and FX bumps rescale the paths exactly and reprice only the trades holding the factor; rate and volatility bumps re-simulate.
- Cost at N = 300: base run 105 s, 78 equity/FX bumps 88 s (about 93x cheaper than re-simulating each), 13 rate/vol re-simulations at 109 s each.
- Faster alternative implemented: pathwise deltas for equity and FX need no repricing (seconds for every name, netting set and date); adjoint differentiation would cover rates and vol too but needs the black-box pricers ported.
- At peak PFE99 (2026-09-20): +1% on all equities moves PFE99 by \$240k; +1bp on USD rates by -\$47k.

The PCA factor model: explainable, but no speed-up

- Factor 1 is the market (20% of variance); factor 2 separates the Tokyo listings from the US names; later factors pick out groups of names. Five factors explain 47% of the variance, ten 62%.
- Speed: none. The correlated-shock step is not faster (1.91 s full against 2.08 s with 5 factors for 5,000 scenarios) and in any case the run is dominated by repricing (about 2134 s).
- Accuracy: netting-set volatilities under 5 factors differ from the full matrix by A +0.0%, B +1.7%, C +3.9%. Sampling error is not consistently lower, because the factor draws add dimensions.
- Decision: keep the full-rank Cholesky as default; the factor model stays as a robustness option.

Data sources

Input	Source	Used for
Trades, pricing library	Capitolis (supplied)	16 trades; all valuation
USD SOFR futures; USD zero curve long end	CME via Databento; Bloomberg snapshot (licensed)	USD curve (our bootstrap plus spliced long end)
Treasury CMT yields 3M-30Y	FRED	USD rate sigma, correlations, G2++, backtest, stress replays
Equities, USDJPY, sectors	yfinance	Spots, vols, correlation, SA-CVA buckets, backtest
SOFR history; credit spreads by rating	FRED (SOFR, ICE BofA OAS)	USD-JPY correlation; CVA credit proxy
JPY OIS history (35 tenors, 2011-2026)	Bloomberg file from the project team	JPY curve, vol, correlations, mean-reversion test
TONA; JGB yields	Bank of Japan API; Japan MoF	JPY rate cross-checks
Swaption cubes, FX forwards	Bloomberg snapshot (licensed)	USD mean reversion, FX forwards
SA-CVA rules; SA-CCR rules	BIS Basel MAR50 (2020); CRE52	Risk weights, correlations, aggregation, EAD

What remains, and what we need from Capitolis

- Data: real counterparty ratings or CDS; credit support annex terms (threshold, minimum transfer amount, initial margin).
- Decisions: which exposure definition prices CVA; whether DVA is recognised; whether a wrong-way-risk model is wanted; SA-CVA versus BA-CVA as the capital reference.
- Longer term: adjoint differentiation (needs the pricers ported), two-factor rates as default (swaption calibration), hybrid sampling for the PCA option, stochastic volatility if option data is provided.
- Assumptions to keep in mind: volatility is a 3-year realised proxy; one static correlation matrix; GBM understates fat tails; own credit and funding spread are assumed; stress tests are instantaneous shocks.
- Ask: are any trades margined, and under what CSA terms?